

Q1. The $E^\ominus$ values for

$\text{Al}^+/\text{Al} = +0.55 \text{ V}$ and $\text{Tl}^+/\text{Tl} = -0.34 \text{ V}$

$\text{Al}^{3+}/\text{Al} = -1.66 \text{ V}$ and $\text{Tl}^{3+}/\text{Tl} = +1.26 \text{ V}$

Identify the incorrect statement

(2023)

- (a) Al is more electropositive than Tl
- (b) Tl^{3+} is a good reducing agent than Tl^{1+}
- (c) Al^+ is unstable in solution
- (d) Al^+ can be easily oxidised Tl than Tl^{3+}

Q2. Molar conductance of an electrolyte increase with dilution according to the equation:

$$\Lambda_m = \Lambda_m^\circ - A\sqrt{c}$$

Which of the following statements are true?

- (a) This equation applies to both strong and weak electrolytes.
- (b) Value of the constant A depends upon the nature of the solvent.
- (c) Value of constant A is same for both BaCl_2 and MgSO_4 .
- (d) Value of constant A is same for both BaCl_2 and $\text{Mg}(\text{OH})_2$.

Choose the most appropriate answer from the options given below: (2023)

- (a) A and B only
- (b) A, B and C only
- (c) B and C only
- (d) B and D only

Q3. The correct value of cell potential in volt for the reaction that occurs when the following two half cells are connected, is

$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s}), E^\circ = -0.44 \text{ V}$

$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$

$E^\circ = +1.33 \text{ V}$

(2023)

- (a) +1.77 V
- (b) +2.65 V
- (c) +0.01 V
- (d) +0.89 V

Q4. The conductivity of centimolar solution of KCl at 25°C is $0.0210 \text{ ohm}^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing the solution at 25°C is 60 ohm. The value of cell constant is-

(2023)

- (a) 3.28 cm^{-1}
- (b) 1.26 cm^{-1}
- (c) 3.34 cm^{-1}
- (d) 1.34 cm^{-1}

Q5. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:



Assertion: In equation $\Delta_r G = -nFE_{\text{cell}}$, value of $\Delta_r G$ depends on n .

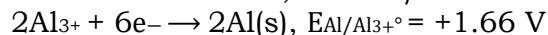
Reason: E_{cell} is an intensive property and $\Delta_r G$ is an extensive property.

In the light of the above statements, choose the correct answer from the options given below:

(2023)

- (a) Both A and R are true but R is NOT the correct explanation of A.
- (b) A is true but R is false.
- (c) A is false but R is true.
- (d) Both A and R are true R R is the correct explanation of A.

Q6. Two half cell reactions are given below:

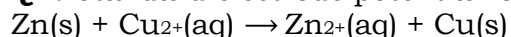


The standard EMF of a cell with feasible redox reaction will be:

(2022)

- (a) -3.47 V
- (b) $+7.09 \text{ V}$
- (c) $+0.15 \text{ V}$
- (d) $+3.47 \text{ V}$

Q7. Standard electrode potential for the cell with cell reaction



is 1.1 V . Calculate the standard Gibbs energy change for the cell reaction.

(Given: $F = 96487 \text{ C mol}^{-1}$)

(2022)

- (a) $-200.27 \text{ J mol}^{-1}$
- (b) $-200.27 \text{ kJ mol}^{-1}$
- (c) $-212.27 \text{ kJ mol}^{-1}$
- (d) $-212.27 \text{ J mol}^{-1}$

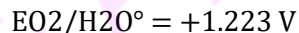
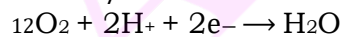
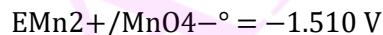
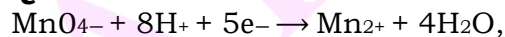
Q8. At 298 K , the standard electrode potentials of Cu^{2+}/Cu , Zn^{2+}/Zn , Fe^{2+}/Fe and Ag^+/Ag are 0.34 V , -0.76 V , -0.44 V and 0.80 V , respectively.

On the basis of standard electrode potential, predict which of the following reaction cannot occur?

(2022)

- (a) $\text{CuSO}_4(\text{aq}) + \text{Zn(s)} \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu(s)}$ (a) The vapour will contain equal amounts of benzene and toluene
- (b) $\text{CuSO}_4(\text{aq}) + \text{Fe(s)} \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu(s)}$
- (c) $\text{FeSO}_4(\text{aq}) + \text{Zn(s)} \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Fe(s)}$
- (d) $2\text{CuSO}_4(\text{aq}) + 2\text{Ag(s)} \rightarrow 2\text{Cu(s)} + \text{Ag}_2\text{SO}_4(\text{aq})$

Q9. Given below are half cell reactions:



will the permanganate ion, MnO_4^- liberate O_2 from water in the presence of an acid?

(2022)

- (a) Yes, because $E_{\text{cell}}^\circ = +0.287 \text{ V}$
- (b) No, because $E_{\text{cell}}^\circ = -0.287 \text{ V}$
- (c) Yes, because $E_{\text{cell}}^\circ = +2.733 \text{ V}$
- (d) No, because $E_{\text{cell}}^\circ = -2.733 \text{ V}$



Q10. Find the emf of the cell in which the following reaction takes place at 298 K
 $\text{Ni(s)} + 2\text{Ag}^+(0.001 \text{ M}) \rightarrow \text{Ni}^{2+}(0.001 \text{ M}) + 2\text{Ag(s)}$

(Given that $E_{\text{cell}}^\circ = 10.5 \text{ V}$, $2.303 \text{ RTF} = 0.059$ at 298 K)

(2022)

- (a) 1.0385 V
- (b) 1.385 V
- (c) 0.9615 V
- (d) 1.05 V

Q11. The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer. **(2021)**

- (a) $390.71 \text{ S cm}^2 \text{ mol}^{-1}$
- (b) $698.28 \text{ S cm}^2 \text{ mol}^{-1}$
- (c) $540.48 \text{ S cm}^2 \text{ mol}^{-1}$
- (d) $201.28 \text{ S cm}^2 \text{ mol}^{-1}$

Q12. The molar conductivity of 0.007 M acetic acid is $20 \text{ S cm}^2 \text{ mol}^{-1}$. What is the dissociation constant of acetic acid?

Choose the correct option.

(2021)

$[\Lambda_{\text{H}^+}^\circ = 350 \text{ S cm}^2 \text{ mol}^{-1}, \Lambda_{\text{CH}_3\text{COO}^-}^\circ = 50 \text{ S cm}^2 \text{ mol}^{-1}]$

- (a) $2.50 \times 10^{-4} \text{ mol L}^{-1}$
- (b) $1.75 \times 10^{-5} \text{ mol L}^{-1}$
- (c) $2.50 \times 10^{-5} \text{ mol L}^{-1}$
- (d) $1.75 \times 10^{-4} \text{ mol L}^{-1}$

Q13. On electrolysis of dil sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be:

- (a) Oxygen gas
- (b) H_2S gas
- (c) SO_2 gas
- (d) Hydrogen gas

Q14. The number of Faradays (F) required to produce 20 g of calcium from molten CaCl_2 (Atomic mass of Ca = 40 g mol^{-1}) is: **(2020)**

- (a) 2
- (b) 3
- (c) 4
- (d) 1

Q15. Identify the reaction from following having top position in EMF series (Std. red. Potential) according to their electrode potential at 298 K. **(2020 Covid Re-NEET)**

- (a) $\text{Fe}^{2+} + 2e^- \rightarrow \text{Fe(s)}$
- (b) $\text{Au}^{3+} + 3e^- \rightarrow \text{Au(s)}$
- (c) $\text{K}^+ + e^- \rightarrow \text{K(s)}$
- (d) $\text{Mg}^{2+} + 2e^- \rightarrow \text{Mg(s)}$

Q16. In a typical fuel cell, the reactants (R) and product (P) are **(2020 Covid Re-NEET)**



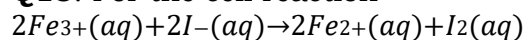
- (a) $R=H_2(g), O_2(g); P=H_2O(l)$
 (b) $R=H_2(g), O_2(g), Cl_2(g); P=HClO_4(aq)$
 (c) $R=H_2(g), N_2(g); P=NH_3(aq)$
 (d) $R=H_2(g), O_2(g); P=H_2O_2(l)$

Q17. For a cell involving one electron $E^\circ_{\text{cell}} = 0.59 \text{ V}$ at 298 K, the equilibrium constant for the cell reaction is: **(2019)**

[Given that $2.303RT/F = 0.059 \text{ V}$ at $T = 298 \text{ K}$]

- (a) 1.0×10^2
 (b) 1.0×10^5
 (c) 1.0×10^{10}
 (d) 1.0×10^{30}

Q18. For the cell reaction



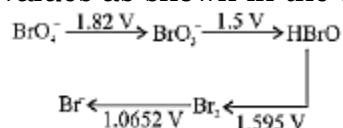
$E_{\text{cell}}^\ominus = 0.24 \text{ V}$ at 298 K. The standard Gibbs energy ($\Delta_r G^\ominus$) of the cell reaction is:

[Given that Faraday constant $F = 96500 \text{ C mol}^{-1}$]

(2019)

- (a) $-46.32 \text{ kJ mol}^{-1}$
 (b) $-23.16 \text{ kJ mol}^{-1}$
 (c) $46.32 \text{ kJ mol}^{-1}$
 (d) $23.16 \text{ kJ mol}^{-1}$

Q19. Consider the change in oxidation state of Bromine corresponding to different emf values as shown in the diagram below:



Then the species undergoing disproportionation is: **(2018)**

- (a) BrO_3^-
 (b) BrO_4^-
 (c) HBrO
 (d) Br_2

Q20. During the electrolysis of molten sodium chloride, the time required to produce 0.10 mol of chlorine gas using a current of 3 amperes is: **(2016-II)**

- (a) 220 minutes
 (b) 330 minutes
 (c) 55 minutes
 (d) 110 minutes

Q21. Zinc can be coated on iron to produce galvanized iron but the reverse is not possible. It is because: **(2016-II)**

- (a) Zinc has lower negative electrode potential than iron
 (b) Zinc has higher negative electrode potential than iron
 (c) Zinc is lighter than iron
 (d) Zinc has lower melting point than iron



Q22. The number of electrons delivered at the cathode during electrolysis by a current of 1 ampere in 60 seconds is: **(2016-II)**

(charge on electron = 1.60×10^{-19} C)

- (a) 3.75×10^{20}
- (b) 7.48×10^{23}
- (c) 6×10^{23}
- (d) 6×10^{20}

Q23. If the E°_{cell} for a given reaction has a negative value, which of the following gives the correct relationships for the values of ΔG° and K_{eq} ? **(2016-II)**

- (a) $\Delta G^\circ < 0; K_{\text{eq}} > 1$
- (b) $\Delta G^\circ < 0; K_{\text{eq}} < 1$
- (c) $\Delta G^\circ > 0; K_{\text{eq}} < 1$
- (d) $\Delta G^\circ > 0; K_{\text{eq}} > 1$

Q24. The molar conductivity of a 0.5 mol dm^{-3} solution of AgNO_3 with electrolytic conductivity of $5.76 \times 10^{-3} \text{ S cm}^{-1}$ at 298 K is: **(2016-II)**

- (a) $0.086 \text{ S cm}^2 \text{ mol}^{-1}$
- (b) $28.8 \text{ S cm}^2 \text{ mol}^{-1}$
- (c) $2.88 \text{ S cm}^2 \text{ mol}^{-1}$
- (d) $11.52 \text{ S cm}^2 \text{ mol}^{-1}$

Q25. The pressure of H_2 required to make the potential of H_2 electrode zero in pure water at 298 K is: **(2016-I)**

- (a) 10^{-4} atm
- (b) 10^{-14} atm
- (c) 10^{-12} atm
- (d) 10^{-10} atm

Q26. A device that converts energy of combustion of fuels like hydrogen and methane, directly into electrical energy is known as: **(2015)**

- (a) Electrolytic cell
- (b) Dynamo
- (c) Ni-Cd cell
- (d) Fuel cell

Q27. The pair of compounds that can exist together is: **(2014)**

- (a) $\text{HgCl}_2, \text{SnCl}_2$
- (b) $\text{FeCl}_2, \text{SnCl}_2$
- (c) FeCl_3, KI
- (d) $\text{FeCl}_3, \text{SnCl}_2$

Q28. Using the Gibbs energy change, $\Delta G^\circ = +63.3 \text{ kJ}$, for the following reaction, $\text{Ag}_2\text{CO}_3(\text{s}) \rightleftharpoons 2\text{Ag}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$ the K_{sp} of $\text{Ag}_2\text{CO}_3(\text{s})$ in water at 25°C is: **(2014)**
($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- (a) 8.0×10^{-12}
- (b) 2.9×10^{-3}
- (c) 7.9×10^{-2}
- (d) 3.2×10^{-26}



Q29. When 0.1 mol MnO_4^{2-} is oxidized, the quantity of electricity required to completely oxidise MnO_4^{2-} to MnO_4^- is: **(2014)**

- (a) 96500 C
- (b) 2×96500 C
- (c) 9650 C
- (d) 96.50 C

Q30. The weight of silver (atomic weight = 108) displaced by a quantity of electricity which displaces 5600 mL of O_2 at STP will be: **(2014)**

- (a) 10.8 g
- (b) 54.0 g
- (c) 108.0 g
- (d) 5.4 g

Q31. At 25°C , molar conductance of 0.1 molar aqueous solution of ammonium hydroxide is $9.54 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and at infinite dilution its molar conductance is $238 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$. The degree of ionization of ammonium hydroxide at the same concentration and temperature is: **(2013)**

- (a) 2.080%
- (b) 20.800%
- (c) 4.008%
- (d) 40.800%

Q32. A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 10 and by passing hydrogen gas around the platinum wire at one atm pressure. The oxidation potential of electrode would be? **(2013)**

- (a) 0.059 V
- (b) 0.59 V
- (c) 0.118 V
- (d) 1.18 V



ANSWER KEY

1. Ans. (b)
2. Ans. (d)
3. Ans. (a)
4. Ans. (b)
5. Ans. (d)
6. Ans. (d)
7. Ans. (c)
8. Ans. (d)
9. Ans. (a)
10. Ans. (a)
11. Ans. (a)
12. Ans. (b)
13. Ans. (a)
14. Ans. (d)
15. Ans. (b)
16. Ans. (a)
17. Ans. (c)
18. Ans. (a)
19. Ans. (c)
20. Ans. (d)
21. Ans. (d)
22. Ans. (b)
23. Ans. (a)
24. Ans. (c)
25. Ans. (d)
26. Ans. (b)
27. Ans. (d)
28. Ans. (b)
29. Ans. (a)

29. Ans. (c)
30. Ans. (c)
31. Ans. (c)
32. Ans. (b)



1. Ans. (b)

Ti_3 act as an oxidising agent not reducing agent.

2. Ans. (d)

B and D statements are correct.

3. Ans. (a)

$$\begin{aligned} E_{\text{cell}}^{\circ} &= E_{\text{C}}^{\circ} - E_{\text{A}}^{\circ} \\ &= (1.33) - (-0.44) \\ &= +1.77 \text{ V} \end{aligned}$$

4. Ans. (b)

Centimolar solution = $1/100 \text{ M} = 0.01 \text{ M}$

Conductivity (k) = $0.0210 \text{ ohm}^{-1} \text{ cm}^{-1}$

Resistance (R) = 60 ohm

$$k = 1/R(l_A)$$

$$\Rightarrow 0.0210 = 1/60(l_A) \Rightarrow l_A = 1.26 \text{ cm}^{-1}$$

5. Ans. (d)

$$\Delta_r G = -nFE_{\text{cell}}$$

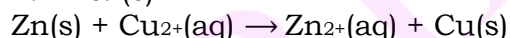
E_{cell} is an intensive property and $\Delta_r G$ is an extensive property as it depends on number of e^- transferred in cell reaction.

6. Ans. (d)

Since, E_{O}° of Al is more than Co^{2+} , so at anode Al will oxidise and at cathode Co^{3+} will reduce.

$$\begin{aligned} E_{\text{cell}}^{\circ} &= (E_{\text{Cathode}}^{\circ})_{\text{RP}} - (E_{\text{Anode}}^{\circ})_{\text{RP}} \\ &= E_{\text{Co}^{3+}/\text{Co}^{2+}}^{\circ} - E_{\text{Al}^{3+}/\text{Al}}^{\circ} \\ &= (1.81) - (-1.66) \\ &= +3.47 \text{ V} \end{aligned}$$

7. Ans. (c)



$$E_{\text{cell}}^{\circ} = 1.1 \text{ V}$$

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ}$$

$$\therefore n = 2$$

$$\Delta G^{\circ} = -2 \times 96487 \times 1.1$$

$$\Delta G^{\circ} = -212271.4 \text{ J mol}^{-1}$$

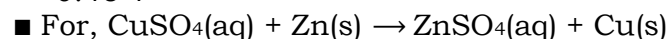
$$\Delta G^{\circ} = -212.27 \text{ kJ mol}^{-1}$$

8. Ans. (d)

For a reaction to be spontaneous, E_{cell}° must be positive. Maximum boiling azeotrope will form

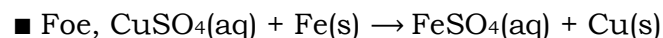
$$\begin{aligned} \blacksquare \text{ For, } \text{FeSO}_4(\text{aq}) + \text{Zn(s)} &\rightarrow \text{ZnSO}_4(\text{aq}) + \text{Fe(s)} \quad E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} \\ &= -0.44 \text{ V} - (-0.76 \text{ V}) \\ &= 0.32 \text{ V} \end{aligned}$$

$$\begin{aligned} \blacksquare \text{ For, } 2\text{CuSO}_4(\text{aq}) + 2\text{Ag(s)} &\rightarrow 2\text{Cu(s)} + \text{Ag}_2\text{SO}_4(\text{aq}) \\ E_{\text{cell}}^{\circ} &= 0.34 \text{ V} - 0.80 \text{ V} \\ &= -0.46 \text{ V} \end{aligned}$$



$$E_{\text{cell}}^{\circ} = 0.34 \text{ V} - (-0.76 \text{ V})$$

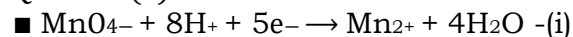
$$= 1.1 \text{ V}$$



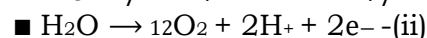
$$E_{\text{cell}}^{\circ} = 0.80 \text{ V} - (-0.44 \text{ V})$$

$$= 1.24 \text{ V}$$

Q9. Ans. (a)

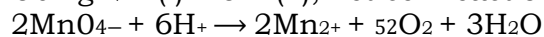


$$E_{\text{MnO}_4^-/\text{Mn}^{2+}}^{\circ} = -E_{\text{Mn}^{2+}/\text{MnO}_4^-}^{\circ} = 1.51 \text{ V}$$



$$E_{\text{O}_2/\text{H}_2\text{O}}^{\circ} = 1.223 \text{ V}$$

Using $2 \times \text{(i)} + 5 \times \text{(ii)}$, net cell reactions is

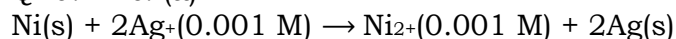


$$E_{\text{cell}}^{\circ} = E_{\text{C}}^{\circ} - E_{\text{A}}^{\circ} = E_{\text{MnO}_4^-/\text{Mn}^{2+}}^{\circ} - E_{\text{O}_2/\text{H}_2\text{O}}^{\circ}$$

$$= 1.51 - 1.223 = 0.287 \text{ V}$$

Since $E_{\text{cell}}^{\circ} > 0$, therefore net cell reaction is spontaneous and so MnO_4^- liberate O_2 from H_2O in presence of an acid.

Q10. Ans. (a)



$$E_{\text{cell}}^{\circ} = 10.5 \text{ V}$$

$$E_{\text{cell}}^{\circ} = E_{\text{cell}}^{\circ} - 0.059n \log[\text{Ni}^{2+}][\text{Ag}^+]^2$$

$$\Rightarrow 10.5 - 0.0592 \log(10^{-3})(10^{-3})^2$$

$$\Rightarrow 10.5 - 0.0592 \log(10)^3$$

$$\Rightarrow 10.5 - 0.0295 \times 3$$

$$\Rightarrow 10.5 - 0.0885$$

11. Ans.(a)

$$\Delta_{\text{NaCl}} = \Delta_{\text{Na}^+} + \Delta_{\text{Cl}^-}$$

$$\Delta_{\text{HCl}} = \Delta_{\text{H}^+} + \Delta_{\text{Cl}^-}$$

$$\Delta_{\text{CH}_3\text{COONa}} = \Delta_{\text{Na}^+} + \Delta_{\text{CH}_3\text{COO}^-}$$

$$\text{Let, } \Delta_{\text{Na}^+} = x, \Delta_{\text{Cl}^-} = y, \Delta_{\text{H}^+} = \Delta_{\text{CH}_3\text{COO}^-} = w$$

Given,

$$x + y = 126.45 \text{ ... (i)}$$

$$y + z = 426.16 \text{ ... (ii)}$$

$$x + w = 91 \text{ ... (iii)}$$

From the above 3 equations, value of

$$z + w = 390.71 \text{ d}$$

12. Ans.(b)

$$\Delta_{\text{M}(\text{CH}_3\text{COOH})}^{\circ} = \Delta_{\text{M}(\text{H}^+)}^{\circ} + \Delta_{\text{M}(\text{CH}_3\text{COO}^-)}^{\circ}$$

$$= 350 + 50 = 400 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\alpha = \Delta_{\text{M}} \Delta_{\text{M}}^{\circ}$$

$$a = 20400 = 5 \times 10^{-2}$$

$$K_a(\text{CH}_3\text{COOH}) = C\alpha^2$$

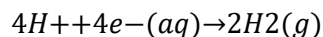
$$= 0.007 \times (5 \times 10^{-2})^2 = 1.75 \times 10^{-5} \text{ mol L}^{-1}$$

13. Ans.(a)

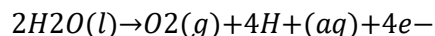


The following reactions take place on electrolysis of dilute sulphuric acid on using Pt electrodes are:

At cathode:



At anode:



Thus, the product obtained at anode is O_2 .

14. Ans.(d)

1 equivalent of any substance is deposited by 1 F of charge.

We have, 20 g calcium

The balanced reaction will

The charge on Ca in $CaCl_2$

Cl has -1 charge so that

$$Ca + 2(-1) = 0$$

$$Ca = 2$$

We have to get Ca from Ca^{2+}

Number of required moles = mass/molar mass

Molar mass of Ca is 40 g/mol and required of Ca is 20 g

Hence number of moles = $20/40$

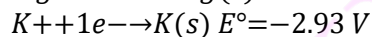
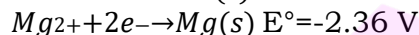
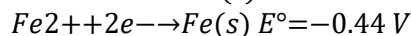
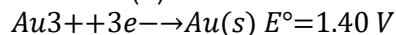
$$= 0.5 \text{ mol}$$

Electricity required to produce 1 mol of calcium = 2F

The electricity required to produce 0.5 mol of calcium = $0.5 \times 2F$

$$= 1F$$

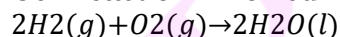
15. Ans.(b)



Au^{3+} occupies the top position in the electrochemical series.

16. Ans.(a)

Cell reaction involved in hydrogen-oxygen fuel cell is



Thus, $R = H_2(g), O_2(g); P = H_2O(l)$

17. Ans.(c)

$$E_{cell} = E^{\circ}_{cell} - 0.059n \log Q \dots (i)$$

At equilibrium, $Q = K_{eq}$ and $E_{cell} = 0$

$$0 = E^{\circ}_{cell} - 0.0591 \log K_{eq} \text{ (from equation (i))}$$

$$\log K_{eq} = \frac{E^{\circ}_{cell}}{0.059} = \frac{0.59}{0.059} = 10$$

$$K_{eq} = 10^{10} = 1 \times 10^{10}$$

18. Ans.(a)

$$\Delta G^{\circ} = -nF E^{\circ}_{cell}$$

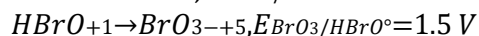
$$= -2 \times 96500 \times 0.24$$

$$= -46320 \text{ J mol}^{-1} = -46320/1000$$

$$= -46.32 \text{ kJ/mol}$$



19. Ans.(c)



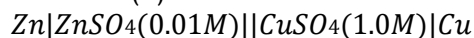
E_{cell}° for the disproportionation of $HBrO$,

$$E_{cell}^\circ = E_{HBrO/Br^-}^\circ - E_{BrO_3^-/HBrO}^\circ$$

$$= 1.595 - 1.5$$

$$= 0.095 V = +ve$$

20. Ans.(d)



$$\therefore E_1 = E_{cell}^\circ - 2.303RT \times F \times \log(0.01)1$$

When concentrations are changed

$$\therefore E_2 = E_{cell}^\circ - 2.303RT \times F \times \log 10.01$$

$$i.e., E_1 > E_2$$

21. Ans.(d)

According to Faraday's first law:

$$w = z \cdot i \cdot t$$

$$z = \frac{E}{96500} (\text{molar mass})$$

$$0.1 \times 71 = 35.5 \times 96500 \times 3 \times t \text{ as } x \text{ factor } 2$$

$$t = 110 \text{ min}$$

22. Ans.(b)

Zn has a higher (-ve) electrode potential that is more reactive than Fe. It is coated on iron substances to provide resistance against rusting such a process is called galvanization. But in reverse, that is Fe cannot be coated on Zn, as corrosion will occur. In above, Zn displaces Fe from its salt solution.

23. Ans.(a)

According to Faraday's law

$$Q = ne$$

$$Q = it$$

$$ne = it$$

$$n = \frac{1 \times 601.6 \times 10^{-19}}{3.75} = 3.75 \times 10^{20} \text{ electrons}$$

24. Ans.(c)

$$\Delta G^\circ = -nFE^\circ_{cell} \quad E^\circ_{cell} = (-ve)$$

$$\text{So, } \Delta G^\circ = (+ve) \quad \Delta G > 0$$

$$\text{Also, } \Delta G^\circ = -2.303 RT \log K_{eq}$$

$$\therefore K_{eq} < 1$$

25. Ans.(d)

$$\text{Concentration} = 0.5 \text{ mol dm}^{-3},$$

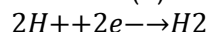
$$K = 5.76 \times 10^{-3} \text{ S cm}^{-1}$$

$$T = 298 \text{ K}$$

$$\lambda_m = K \times 1000M$$

$$= 5.76 \times 10^{-3} \times 0.5 = 11.52 \text{ S cm}^2 \text{ mol}^{-1}$$

26. Ans.(b)



$$\begin{aligned} E &= E^\circ - 0.0592 \log_{PH_2(H+)^2} \\ 0 &= 0 - 0.0592 \log_{PH_2(10^{-7})^2} \\ \log 1 &= 0 \\ PH_2 &= (10^{-7})^2 = 10^{-14} \text{ atm.} \end{aligned}$$

27. Ans.(d)

28. Ans.(b)

Sn^{2+} cannot reduce Fe^{2+} , so FeCl_2 and SnCl_2 can exist together.

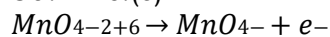
29. Ans.(a)

$$\Delta G^\circ = -2.303 RT \log K_{sp}$$

$$63300 = -2.303 \times 8.314 \times 298 \log K_{sp}$$

$$K_{sp} \sim 8 \times 10^{-12}$$

30. Ans.(c)



0.1 mole 0.1 mole

$$Q = nF = 0.1 \times 96500 = 9650 \text{ C}$$

31. Ans.(c)

$$\begin{aligned} W_{O_2} &= 560022400 \times 32 = 8g = 1 \text{ equivalent} \\ &= 1 \text{ equivalent of Ag} \\ &= 108 g \end{aligned}$$

32. Ans.(c)

According to Kohlrausch's law of limiting molar conductivity:

$$\alpha = \Lambda_m \Lambda_{m^*} = 9.54238 = 0.04008$$

% Dissociation (□) = 4.008%